

$$f : \mathbb{R}^3 \rightarrow \mathbb{R}$$

$$\begin{aligned}\max f(x) &= x_1 + 2x_2 + x_3 \\ x_1 + 2x_2 - x_3 &= 4 \\ -x_1 + x_2 + x_3 &= 8 \\ x_1 + 5x_2 + x_3 &= 16 \\ x_1, x_2, x_3 &\geq 0\end{aligned}$$

Metodo delle due fasi:

Fase I

$$\begin{aligned}\min f(x) &= a_1 + a_2 + a_3 \\ x_1 + 2x_2 - x_3 + a_1 &= 4 \\ -x_1 + x_2 + x_3 + a_2 &= 8 \\ x_1 + 5x_2 + x_3 + a_3 &= 16 \\ x_1, x_2, x_3, a_1, a_2, a_3 &\geq 0\end{aligned}$$

Calcolo dei costi:

$$c_i = c_N + c_B^T \cdot B_N^{-1}$$

$$c_1 = 0 - (1; 1; 1) \cdot \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} = -1$$

$$c_2 = 0 - (1; 1; 1) \cdot \begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix} = -8$$

$$c_3 = 0 - (1; 1; 1) \cdot \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} = -1$$

$$c_4 = 1 - (1; 1; 1) \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = 0$$

$$c_5 = 1 - (1; 1; 1) \cdot \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = 0$$

$$c_6 = 1 - (1; 1; 1) \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = 0$$

$$c_7 = 0 - (1; 1; 1) \cdot \begin{pmatrix} 4 \\ 8 \\ 16 \end{pmatrix} = -28$$

Metodo del simplesso:

	x_1	x_2	x_3	a_1	a_2	a_3	b
a_1	1	2	-1	1	0	0	4
a_2	-1	1	1	0	1	0	8
a_3	1	5	1	0	0	1	16
	-1	-8	-1	0	0	0	-28

Entra x_2 ed esce a_1 :

Forma Standard

$$\begin{aligned}\min f(x) &= -x_1 - 2x_2 - x_3 \\ x_1 + 2x_2 - x_3 &= 4 \\ -x_1 + x_2 + x_3 &= 8 \\ x_1 + 5x_2 + x_3 &= 16 \\ x_1, x_2, x_3 &\geq 0\end{aligned}$$

	x_1	x_2	x_3	a_1	a_2	a_3	b
a_1	1	2	-1	1	0	0	4
a_2	-1	1	1	0	1	0	8
a_3	1	5	1	0	0	1	16
	-1	-8	-1	0	0	0	-28

$$R1 \leftarrow R1 \cdot \frac{1}{2}$$

	x_1	x_2	x_3	a_1	a_2	a_3	b
a_1	$\frac{1}{2}$	1	$-\frac{1}{2}$	$\frac{1}{2}$	0	0	2
a_2	-1	1	1	0	1	0	8
a_3	1	5	1	0	0	1	16
	-1	-8	-1	0	0	0	-28

$$R2 \leftarrow R2 - R1$$

$$\begin{array}{cccccc|c} -1 & 1 & 1 & 0 & 1 & 0 & 8 \\ \frac{1}{2} & 1 & -\frac{1}{2} & \frac{1}{2} & 0 & 0 & 2 \\ \hline -\frac{3}{2} & 0 & \frac{3}{2} & -\frac{1}{2} & 1 & 0 & 6 \end{array}$$

↓

	x_1	x_2	x_3	a_1	a_2	a_3	b
a_1	$\frac{1}{2}$	1	$-\frac{1}{2}$	$\frac{1}{2}$	0	0	2
a_2	$-\frac{3}{2}$	0	$\frac{3}{2}$	$-\frac{1}{2}$	1	0	6
a_3	1	5	1	0	0	1	16
	-1	-8	-1	0	0	0	-28

$$R3 \leftarrow R3 - R1 \cdot 5$$

$$\begin{array}{cccccc|c} 1 & 5 & 1 & 0 & 0 & 1 & 16 \\ \frac{5}{2} & 5 & -\frac{5}{2} & \frac{5}{2} & 0 & 0 & 10 \\ \hline -\frac{3}{2} & 0 & \frac{7}{2} & -\frac{5}{2} & 0 & 1 & 6 \end{array}$$

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	x_1	x_2	x_3	a_1	a_2	a_3	b
a_1	$\frac{1}{2}$	1	$-\frac{1}{2}$	$\frac{1}{2}$	0	0	2
a_2	$-\frac{3}{2}$	0	$\frac{3}{2}$	$-\frac{1}{2}$	1	0	6
a_3	$-\frac{3}{2}$	0	$\frac{7}{2}$	$-\frac{5}{2}$	0	1	6
	-1	-8	-1	0	0	0	-28

$$R4 \leftarrow R4 + R1 \cdot 8$$

$$\begin{array}{cccccc|c} -1 & -8 & -1 & 0 & 0 & 0 & -28 \\ 4 & 8 & -4 & 4 & 0 & 0 & 16 \\ \hline 3 & 0 & -5 & 4 & 0 & 0 & -12 \end{array}$$

↓

	x_1	x_2	x_3	a_1	a_2	a_3	b
x_2	$\frac{1}{2}$	1	$-\frac{1}{2}$	$\frac{1}{2}$	0	0	2
a_2	$-\frac{3}{2}$	0	$\frac{3}{2}$	$-\frac{1}{2}$	1	0	6
a_3	$-\frac{3}{2}$	0	$\frac{7}{2}$	$-\frac{5}{2}$	0	1	6
	3	0	-5	4	0	0	-12

Entra x_3 ed esce a_3 :

	x_1	x_2	x_3	a_1	a_2	a_3	b
x_2	$\frac{1}{2}$	1	$-\frac{1}{2}$	$\frac{1}{2}$	0	0	2
a_2	$-\frac{3}{2}$	0	$\frac{3}{2}$	$-\frac{1}{2}$	1	0	6
a_3	$-\frac{3}{2}$	0	$\frac{7}{2}$	$-\frac{5}{2}$	0	1	6
	3	0	-5	4	0	0	-12

$$R3 \leftarrow R3 \cdot \frac{2}{7}$$

	x_1	x_2	x_3	a_1	a_2	a_3	b
x_2	$\frac{1}{2}$	1	$-\frac{1}{2}$	$\frac{1}{2}$	0	0	2
a_2	$-\frac{3}{2}$	0	$\frac{3}{2}$	$-\frac{1}{2}$	1	0	6
a_3	$-\frac{5}{7}$	0	1	$-\frac{5}{7}$	0	1	$\frac{12}{7}$
	3	0	-5	4	0	0	-12

$$R1 \leftarrow R1 + R3 \cdot \frac{1}{2}$$

$$\begin{array}{cccccc|c} \frac{1}{2} & 1 & -\frac{1}{2} & \frac{1}{2} & 0 & 0 & 2 \\ -\frac{3}{2} & 0 & \frac{3}{2} & -\frac{1}{2} & 0 & \frac{1}{2} & \frac{6}{7} \\ \hline \frac{3}{7} & 1 & 0 & \frac{1}{7} & 0 & \frac{1}{2} & \frac{20}{7} \end{array}$$

↓

	x_1	x_2	x_3	a_1	a_2	a_3	b
x_2	$\frac{2}{7}$	1	0	$\frac{1}{7}$	0	$\frac{1}{2}$	$\frac{20}{7}$
a_2	$-\frac{3}{2}$	0	$\frac{3}{2}$	$-\frac{1}{2}$	1	0	6
a_3	$-\frac{5}{7}$	0	1	$-\frac{5}{7}$	0	1	$\frac{12}{7}$
	3	0	-5	4	0	0	-12

$$R2 \leftarrow R2 - R3 \cdot \frac{3}{2}$$

$$\begin{array}{cccccc|c} -\frac{3}{2} & 0 & \frac{3}{2} & -\frac{1}{2} & 1 & 0 & 6 \\ -\frac{9}{2} & 0 & \frac{3}{2} & -\frac{15}{2} & 0 & \frac{3}{2} & \frac{18}{7} \\ \hline -\frac{14}{7} & 0 & 0 & \frac{4}{7} & 1 & -\frac{3}{2} & \frac{24}{7} \end{array}$$

↓

	x_1	x_2	x_3	a_1	a_2	a_3	b
x_2	$\frac{2}{7}$	1	0	$\frac{1}{7}$	0	$\frac{1}{2}$	$\frac{20}{7}$
a_2	$\frac{6}{7}$	0	0	$\frac{4}{7}$	1	$-\frac{3}{2}$	$\frac{24}{7}$
a_3	$-\frac{3}{7}$	0	1	$-\frac{5}{7}$	0	1	$\frac{12}{7}$
	3	0	-5	4	0	0	-12

$$R4 \leftarrow R4 + R3 \cdot 5$$

$$\begin{array}{cccccc|c} 3 & 0 & -5 & 4 & 0 & 0 & -12 \\ -\frac{15}{7} & 0 & 5 & -\frac{25}{7} & 0 & 5 & \frac{60}{7} \\ \hline \frac{6}{7} & 0 & 0 & \frac{3}{7} & 0 & 5 & -\frac{24}{7} \end{array}$$

↓

	x_1	x_2	x_3	a_1	a_2	a_3	b
x_2	$\frac{2}{7}$	1	0	$\frac{1}{7}$	0	$\frac{1}{2}$	$\frac{20}{7}$
a_2	$\frac{6}{7}$	0	0	$\frac{4}{7}$	1	$-\frac{3}{2}$	$\frac{24}{7}$
x_3	$-\frac{3}{7}$	0	1	$-\frac{5}{7}$	0	1	$\frac{12}{7}$
	$\frac{6}{7}$	0	0	$\frac{3}{7}$	0	5	$-\frac{24}{7}$

Portando i vincoli in forma standard con le variabili artificiali a_1, a_2 e a_3 e risolvendo la **Fase I** ($\min f(x) = a_1 + a_2 + a_3$), si ottiene un valore ottimo **strettamente positivo** ($\min f(x) = \frac{24}{7} > 0$). Per il metodo delle due fasi, $\min f(x) > 0 \Rightarrow$ **Non esiste nessuna soluzione ottima**.